

Software Requirements Specification

Personalized Meal & Diet Subscription Manager

Project SRS based on the completed requirements, use cases, RTM, and architecture deliverables.

1. Introduction

1.1 Purpose

This Software Requirements Specification defines the functional and non-functional requirements for a clinical dietary subscription service that generates weekly meal menus according to subscriber dietary constraints and supports subscription and delivery fulfillment tracking.

1.2 Scope

The system manages subscriber dietary profiles, generates weekly meal plans using metabolic and allergen constraints, provides dietitian review and approval, supports subscription schedule changes, and records daily meal delivery fulfillment status.

1.3 Actors

Actor	Role
Subscriber	Manages dietary information, views meal plans, and manages subscription scheduling.
Dietitian	Reviews and approves generated meal plans.

2. System Overview

The system follows a layered application architecture with Presentation, Business/Application, and Data layers. The Business/Application layer contains services for dietary profile management, meal-plan generation, dietitian review and approval, authentication and authorization, subscription management, and delivery fulfillment. The Data layer contains the application database.

3. Functional Requirements

ID	Requirement
FR-001	Generate weekly meal plans according to macro-nutrient constraints and allergen exclusions.
FR-002	Allow subscriber to manage dietary profile.
FR-003	Allow dietitian to review and approve generated meal plans.
FR-004	Allow subscriber to pause, skip, or reschedule subscription.
FR-005	Record and display daily meal delivery fulfillment status.

4. Non-Functional Requirements

ID	Requirement
NFR-001	Support subscription schedule modifications up to 12 hours before dispatch.
NFR-002	Protect subscriber dietary information using authenticated access and role-based authorization.

5. Use Cases

Use Case	Description
UC-01 Manage Dietary Profile	Subscriber manages the dietary information used by the meal planning process.
UC-02 Generate Weekly Meal Plan	System generates a weekly plan according to the saved dietary profile and active subscription.
UC-03 Review & Approve Meal Plan	Dietitian reviews a generated plan and approves it for availability.
UC-04 Manage Subscription	Subscriber pauses, skips, or reschedules the subscription.
UC-05 Track Delivery Fulfillment	System records and displays daily meal delivery fulfillment status.

6. Detailed Use Case: UC-02 Generate Weekly Meal Plan

Preconditions: The subscriber is authenticated, has a saved dietary profile, and has an active subscription for the requested week.

Postconditions: A weekly plan is generated according to constraints, saved for dietitian review, and contains no excluded allergen.

Main Flow: (1) Subscriber selects Generate Weekly Meal Plan. (2) System retrieves the dietary profile and subscription. (3) System validates calorie, macro-nutrient, and allergen constraints. (4) System generates a menu within the allowed limits. (5) System checks the generated plan. (6) System saves it as Pending Dietitian Review. (7) Dietitian is notified. (8) Dietitian reviews and approves the plan. (9) The approved plan becomes available to the subscriber.

Alternate Flow – Constraint Violation: If a generated plan violates a constraint, the system identifies the violation and attempts regeneration. If a valid plan cannot be generated, the plan is flagged for dietitian intervention.

7. System Interfaces / Major Components

Component	Responsibility
Subscriber Interface	Provides subscriber-facing functions.
Dietitian Interface	Provides dietitian review and approval functions.
Dietary Profile Management Service	Maintains subscriber dietary profiles.
Meal Plan Generation Service	Generates plans using dietary and subscription constraints.
Dietitian Review & Approval Service	Handles review and approval of generated plans.
Authentication & Authorization Service	Provides authenticated access and role-based authorization.
Subscription Management Service	Handles pause, skip, and reschedule actions.
Delivery Fulfillment Service	Records and displays daily delivery fulfillment status.
Application Database	Stores application data.

8. Constraints and Assumptions

The requirements specify Subscriber and Dietitian as the principal actors. Subscription modifications are subject to the 12-hour-before-dispatch constraint. Access to dietary information requires authentication and role-based authorization. The architecture uses a layered organization; the requirements do not mandate a specific web or mobile platform.

9. Traceability

Requirement	Mapped Use Case
FR-001	UC-02 Generate Weekly Meal Plan
FR-002	UC-01 Manage Dietary Profile

Requirement	Mapped Use Case
FR-003	UC-03 Review & Approve Meal Plan
FR-004	UC-04 Manage Subscription
FR-005	UC-05 Track Delivery Fulfillment
NFR-001	UC-04 Manage Subscription
NFR-002	UC-01 to UC-05 (access control applies to system functions)

10. Acceptance Summary

The system is considered requirements-complete when all five functional requirements and two non-functional requirements are implemented and can be traced to the corresponding use cases and verification/acceptance criteria in the RTM.